

SUMMARY

AI Engineer with hands-on experience developing and operationalizing end-to-end AI solutions across classical ML, deep learning, and generative AI, covering everything from data preparation to production deployment of RAG pipelines, LLM-powered applications, and model inference systems.

TECHNICAL SKILLS

Programming: Python, SQL, C/C++, Java, JavaScript, Bash
ML & AI: PyTorch, Scikit-learn, Pandas, NumPy, OpenCV
GenAI & LLMs: RAG, Vector DBs, Prompt Engineering
Databases: MySQL, PostgreSQL, SQLite, MongoDB
DevOps & Tools: Docker, FastAPI, REST APIs, Git, CI/CD

EXPERIENCE

- AI Research Assistant**
Eötvös Loránd University (ELTE)
• Build and optimize DP-SGD federated learning pipelines for privacy-preserving synthetic data generation ($\epsilon = 1$), engineering data preparation and feature workflows to benchmark privacy-utility trade-offs
• Author and teach the open-source IDaSec Practicum (8 labs), documenting AI architectures and best practices in responsible AI
• Collaborate with cross-functional teams and mentor graduate students in AI engineering and MLOps workflows
- Data Analyst**
Djezzy
• Analyzed subscriber telecom datasets using Python and SQL to identify usage trends and customer behavior patterns
• Built automated data preparation and feature engineering pipelines, facilitating data-driven decision making and reporting
• Collaborated with analytics and cross-functional teams to model subscriber metrics and optimize customer retention strategies

PROJECTS

- CASPER**
Python, Differential Privacy, RAG, T5, MedCPT, Docker
• Built a privacy-preserving RAG framework that clusters sensitive medical records with DP-KMeans and inverts noisy centroids into synthetic prototypes, enabling LLMs to retrieve on private data without accessing raw records
• Achieved technical Recall@10 of 0.91 at $\epsilon \approx 44.28$, $\delta = 10^{-5}$, with 50% MIA AUC (random-guessing), demonstrating formal and empirical privacy guarantees
- Context-Aware Access Control for IoT Device**
FastAPI, Python, Multi-Agent LLMs, Docker, Wazuh, Prometheus
• Built a hybrid zero-trust IoT authorization framework combining a deterministic validation pre-filter (< 1 ms fast path) with multi-agent LLM reasoning (97 ms slow path) for autonomous context-aware access control
• Achieved 94.1% decision accuracy vs. 82.4% for the RBAC baseline and blocked 100% of 105 red-team adversarial attacks, with fail-secure DENY defaults on LLM rate limits or timeouts
- MyVellum**
FastAPI, Groq API, LLM Orchestration, RAG, Docker, CI/CD
• Architected an LLM-powered app with modular prompt engineering, vector-backed RAG pipelines, and retrieval optimization
• Built automated evaluation gates into CI/CD for relevance and quality metrics; integrated Groq API and FastAPI with Docker

EDUCATION

- PhD in Informatics (GPA: 5/5)**
Eötvös Loránd University (ELTE)
Thesis: Applied synthetic data generation for secure, privacy-preserving AI systems (Supervised by Pr. Lendák Imre)
- M.Sc. in Artificial Intelligence – Ranked 1st in Class (GPA: 3.7/4.0)**
University of Khenchela
Thesis: Hair Dermoscopic Image Segmentation Using Deep Learning (Supervised by Pr. Dalal Bardou)
- B.Sc. in Mathematics and Computer Science – Ranked 1st in Class (GPA: 3.9/4.0)**
University of Khenchela
Final Year Project: A learning management system for Algerian students.

AWARDS & CERTIFICATIONS

- Stipendium Hungaricum Scholarship:** Fully funded PhD studies at ELTE
2nd Place — Starthon'25 Hackathon: University of El Oued, Algeria
Software Engineering (Back-end Specialization): ALX Africa
AI Career Essentials (AiCE): ALX Africa
- Sep. 2025
Apr. 2025
Feb. 2025
Jul. 2024